

LIZHONG CHEN

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Education

- The University of Melbourne** Doctor of Philosophy (PhD) in Statistics 02/2017 – 02/2022
Supervisors: A/Prof. Guoqi Qian, Prof. Yuriy Kuleshov, Dr Tingjin Chu
Thesis: Model selection and averaging by Gibbs sampler with a tropical cyclone seasonal forecasting application
- Peking University** Master of Science (M.S.) in Mathematics 09/2013 – 07/2016
Mentor: A/Prof. Houhong Fan
Thesis: A Report on Computations of Cohomology Rings and Homotopy Groups of Lie Groups
- Peking University** Bachelor of Science (B.S.) in Mathematics 09/2013 – 07/2016

Research Experience

- Smyth Lab, WEHI** *Research Officer* 03/2022 – present
Supervisor: Prof. Gordon K Smyth

Publications

- [1] Jianan Wang, Lizhong Chen, Daniel V Brown, Chris Y Chiu, and Terence P Speed. CMDdemux: an efficient single cell demultiplexing method. *Nucleic Acids Research*, 54(8), 2026.
- [2] Pedro L Baldoni#, Lizhong Chen#, Mengbo Li, Yunshun Chen, and Gordon K Smyth. Dividing out quantification uncertainty enables assessment of differential transcript usage with limma and edgeR. *Nucleic Acids Research*, 53(22), 2025.
- [3] Yunshun Chen, Lizhong Chen, Aaron TL Lun, Pedro L Baldoni, and Gordon K Smyth. edgeR v4: powerful differential analysis of sequencing data with expanded functionality and improved support for small counts and larger datasets. *Nucleic Acids Research*, 53(2), 2025.
- [4] Pedro L Baldoni, Lizhong Chen, and Gordon K Smyth. Faster and more accurate assessment of differential transcript expression with Gibbs sampling and edgeR v4. *NAR Genomics and Bioinformatics*, 6(4), 2024.
- [5] Guoqi Qian#, Lizhong Chen#, and Yuriy Kuleshov. Improving methodology for tropical cyclone seasonal forecasting in the Australian and the South Pacific Ocean regions by selecting and averaging models via Metropolis–Gibbs sampling. *Remote Sensing*, 14(22), 2022.

contribute equally.

Preprints

- [1] Jianan Wang, Lizhong Chen, Rachel Thijssen, Belinda Phipson, and Terence P Speed. GLMsim: a GLM-based single cell RNA-seq simulator incorporating batch and biological effects. *bioRxiv*, 2024.

Talks and Presentations

- edgeR quasi-likelihood: correcting deviance for bias** Bioinformatics division seminar, WEHI 08/2022
- Extending edgeR for small counts and large samples** Bioinformatics division seminar, WEHI 05/2023
- Applying edgeR for alternative splicing analysis** Bioinformatics division seminar, WEHI 04/2024
- edgeRv4 with expanded functionality and improved support** Short talk, EuroBioC2025 09/2025
- Analyzing paired count data using edgeR** Bioinformatics division seminar, WEHI 09/2025

Teaching Experience

- The University of Melbourne** *Tutor* 07/2017 - 02/2022
Linear Algebra, Accelerated Mathematics I/II, Complex Analysis, Multivariate Statistical Techniques
- The University of Melbourne** *Assignment and Exam Marker* 07/2018 - 07/2021
Statistical Modelling, Mathematical statistics
- Peking University** *Teaching assistant* 09/2013 - 07/2016
Mathematical analysis I/II, Calculus I, Advanced algebra I

Honors and scholarships

Melbourne Research Scholarship	The University of Melbourne	2017-2021
Outstanding Graduate of Beijing		2016
Outstanding Graduate of Peking University	Peking University	2016
Guanghua Scholarship	Peking University	2015
National Scholarship	Peking University	2014
Second-Class Freshmen Scholarship	Peking University	2009-2012

Software Projects

IBGS: Iterated Blockwise Gibbs Sampler

IBGS is a CRAN package to perform variable selection for generalized linear models and the Cox proportional-hazards model when the number of predictors is very large. The sampler is implemented in C with parallel block screening through OpenMP, and returns a small set of high-scoring models together with marginal inclusion probabilities and model-averaged predictions.

edgeR: Empirical Analysis of Digital Gene Expression Data in R

edgeR is a Bioconductor R package to perform differential expression analysis for the sequencing data like RNA-seq, ChIP-seq and ATAC-seq. In edgeR v4, I develop the new quasi-likelihood method based on the quadratic mean-variance relationship that we can isolate the biological and technical variation. I improve the quasi-dispersion estimation by the adjusted deviance statistics and support lowly expressed counts data. Besides, I update the computational code from C++ to C for simple structure.

statmod: Statistical Modeling

statmod is a CRAN R package that is a collection of statistical tools. I introduce a new function `expectedDeviance` to provide an approximation to the expected unit deviance and degree of freedom for Poisson, Binomial, and Negative Binomial distribution, which corrects the bias of chi-square distribution approximation of 1 degree of freedom.

limma: Linear Models for Microarray and Omics Data

limma is a Bioconductor R package to perform differential expression analysis for the microarray, sequencing and omics data. I update the estimation of the prior distribution in empirical Bayes approach by a two-steps method, combining the local moment estimates and profile likelihood estimates. It improves the performance when the observed degrees of freedom from the models are unequal and relatively small.

fastLISA: Fast Local Indicators of Spatial Association

fastLISA is a CRAN package to compute seven families of LISA statistics with a plain-C backend, optional OpenMP multi-threading, and a modern xoshiro256 random number generator for permutation-based inference. It accepts any `spdep` listw spatial weights object including custom and non-contiguity (e.g. distance-decay) weights and returns compact, inspectable, `spdep`-compatible matrices.

Supervision

Hasini Herath	<i>PhD (Medical Biology), University of Melbourne</i>	2025 –
Co-supervised with Pedro Baldoni and Gordon Smyth		
Xueming Li	<i>PhD (Mathematics and Statistics), University of Melbourne</i>	2024 –
Co-supervised with Guoqi Qian		
Lei Qin	<i>PhD (Medical Biology), University of Melbourne</i>	2024 –
Co-supervised with Yunshun Chen and Gordon Smyth		
Shama Deb	<i>MSc (Mathematics and Statistics), University of Melbourne</i>	2025 – 2026
Co-supervised with Gordon Smyth		
Yuchen Liu	<i>MSc (Mathematics and Statistics), University of Melbourne</i>	2025 – 2026
Co-supervised with Tingjin Chu		

Referee services

Applied mathematical modelling, Scientific reports, Nucleic Acids Research, Australian & New Zealand Journal of Statistics